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Data Analytics (2026-HS)

General information

Lecturers

- Dr. Mario Gellrich (module responsible)
- Dr. Alexander Walter

Documents

- All course materials can be found on the electronic learning platform "Moodle" and on GitHub.

Tools and software

- Generally, we will use GitHub Codespaces, Python and Jupyter notebooks for all exercises.
- Installation instructions for additional tools will be provided or linked on Moodle.

Location / teaching format

Large classes (German: Grossklassen, GK) are organized as follows:

- All lectures will be held on-site in Winterthur (see PrePlan for rooms and schedules).

Small classes (German: Kleinklassen, KK) are organized as follows:

- In the first week of the semester, all small classes will be held on-site in Winterthur (see WebUntis for rooms and schedules).
- From the second week of the semester onwards, small classes will take place online via MS Teams.
- Note that all online small classes can be attended by every student at any time! The times for the small classes are:
  - Wednesday 19:45 – 21:20
  - Thursday 12:00 – 13:45
  - Thursday 14:00 – 15:45
  - Thursday 18:00 – 19:35
  - Thursday 19:45 – 21:20
- Questions regarding the small classes: please write an email to walx@zhaw.ch

Changes in the location and form of instruction will be announced in a timely manner.

Weekly exercises

The weekly exercises in the small classes must **not** be submitted on Moodle; they serve as preparation for the written exam.

End of module exam / final grade

Only the current module description is relevant with regard to the exam and composition of the final grade.

The final grade is determined as:

- 60% grade of written exam (regular end-of-module exam, MEP)
- 40% grade of project work with the two parts (each 20%):
  - group presentation of the project concept (for details see below)
  - final project submission on Moodle (for details see below)

The written exam will …

- last 60 minutes
- be open book
- cover the entire material of the module
- contain single-choice questions (4 possible answers per question, only one is correct).

Announcements

Help & Heureka!!! forum for knowledge exchange and questions Mark as done

Link to register for the project work and find the date/time of the project concept presentation Mark as done

Note that the date/time of the presentation will initially be defined by the lecturer once all groups are registered!

ModuleDescription w.BA.XX.3DA-WIN.XX

Semesterprogram w BA XX DA HS26

Junior Data Analyst - JobCloud AG

Data Science Winter School Feb 2027

Mark as done

Programming environment & recommended subscriptions

In this course, we will mainly work with Python and Visual Studio Code using GitHub Codespaces.

Therefore, you need a **GitHub Pro account**, which you can apply for via 'educational benefits' on GitHub. This web page describes how to do it.

After you have created your own GitHub Pro account, you can create a Codespace from this repository:

https://github.com/mario-gellrich-zhaw/data_analytics.git
We recommend setting up a payment method and limiting the amount to **10 U.S. dollars per month**.

Although AI models (LLMs) are part of the GitHub ecosystem, the number of AI credits per month is limited. We therefore additionally recommend purchasing either a **Copilot Pro** or a **Claude Pro** subscription. Both can be used from within VS Code via extensions (details will be explained in class).

Mark as done

Evaluation of the project concept presentation

Presentation Outline:

- The ***presentations*** **of the project concept** will be on-site in Winterthur and take place at the beginning of the lectures (large classes) from week 10 onwards (3-4 groups per lecture, 10 min. per group).
- For the presentation, provide a slide set (PDF-Document) and upload it to Moodle (see link below).
- A link to the presentation dates/times is available on Moodle (same link as for the group registration).
- Note that the dates/times of the presentations will determined by the lecturer once the project groups are registered.

To be evaluated:

1. **Structure & outline** (Clear outline presented and followed? Logical flow with recognizable thread? Balanced weighting of parts? Title clear? Names of all students available?)
2. **Background & problem** (Context of the project clear? Problem convincingly motivated?)
3. **Objective & research question** (Clearly defined?)
4. **Data** (Source shown? Structure of the data explained?)
5. **Methods** (Methods appropriate for the research question? Correctly described and justified?)
6. **Expected results** (Plausible? Linked back to the research question?)
7. **Language** (Freely spoken?)
8. **Quality of the presentation** (Quality of graphics, tables & slides according to scientific standards?)

Grading scale (criteria 1–8)

Grade	Description
6	**Excellent** — the criterion is fulfilled completely and convincingly, without notable weaknesses
5	**Good** — the criterion is fulfilled well, with only minor weaknesses
4	**Sufficient** — the criterion is fulfilled in its essential aspects; basic requirements are met
3	**Insufficient** — key elements of the criterion are missing or incorrect
2	**Poor** — the criterion is addressed only in fragments
1	**Not fulfilled** — the criterion is not addressed / no performance shown

Each of the criteria 1–8 is graded independently on the scale 1–6. The final grade for the presentation of the project concept is the average of these eight grades.

Submission of the presentation (PDF-document) of the project concept (Deadline: January 10, 2027, 11:59 p.m.)
Opened: Monday, 14 September 2026, 12:00 AM **Due:** Sunday, 10 January 2027, 11:59 PM

Mark as done

Evaluation of the final project (max. 22 points)

Minimum requirements (9 points)

The project must be a Data Analytics project, i.e., must include content of the Data Analytics module.

The project must integrate several topics presented in the module, at a minimum these are the following:

1. **Data collection** using Web Scraping and/or a Web API. The data must be collected by the group itself; ready-made datasets (e.g. from Kaggle) are not permitted as the primary data source.
2. **Data preparation and integration:** the project must combine data from at least two different sources (e.g. scraped data enriched with data from opendataswiss). The integration step (matching keys, joining, resolving inconsistencies, number of records lost/gained) must be documented.
3. **Data storage** in a SQLite database including SQL queries from within Python.
4. **Rich non-graphical and graphical exploratory data analysis (EDA).**
5. **Modeling:** use of either regression or classification. The research question must go beyond a standard prediction task (e.g. include a comparison between groups, regions, time periods, or data sources).
6. **Model evaluation** using suitable measures of fit (e.g. r-squared, RMSE, accuracy, recall, precision, ...).
7. **Correct interpretation** of model results and measures of fit.
8. **Documentation of AI usage** for creating the code and the presentation (the use of AI is not only allowed but desired), including the advantages and disadvantages you found and where something did not work with the help of AI.
9. **Making the material** (data, Jupyter notebooks, ...) available on Moodle.

Note: The topic will be formally approved by the lecturers at the presentation of the project concept on-site in Winterthur (week 10 onwards). If you

are unsure whether your topic meets the requirements and can achieve the points for the overall project (see the details on Moodle), please contact the lecturers **before you prepare the project concept**. A short email with your topic and research question (2–3 sentences) is enough, and you will get an initial assessment. Overused standard topics (e.g. plain used-car or real-estate price prediction without an extended research question) will be graded lower on creativity.

Additional points, if the following criteria are met (max. 6 points)

1. **Creativity of implementation** (creative is anything not specified in the lessons and exercises).
2. **Use of a PostgreSQL or similar database** (no SQLite!) with, e.g. SQL joins and aggregations from within Python.
3. **Integration and visualization of geographical data.**
4. **Use of a Chi-squared test, analysis of variance (ANOVA), or correlation analysis** (each test must include a p-value).
5. **Use of k-means clustering** in addition to the regression or classification model.
6. **Code on a publicly available GitHub repository** (provide the GitHub project URL in the project presentation).

Project presentation in the form of a video recording (max. 7 points)

To be evaluated:

1. **Structure & outline**
2. **Topic presentation** (Topic clearly defined? How convincingly presented?)
3. **Content** (substance, depth, weighting of parts)
4. **Language / auxiliary tools** (Freely spoken?)
5. **Quality of the presentation & slides**
6. **Reflection on AI usage:** the presentation (slides and video) must include which AI tools were used and how, the advantages and disadvantages you found when using AI, and where something did not work with the help of AI.
7. **Time keeping:** length of presentation in minutes = number of students per group × 5 minutes.

Notes:

→ Use the slide set (PDF) as the basis for the presentation.

→ Show the Python code & live demos only if it improves the understanding of methods and results.

→ Do not just scroll through the code in the Jupyter notebooks without explaining the code and results.

→ AI is a tool, not a substitute for understanding. The evaluation focuses on your ability to explain and justify your data, methods, and results.

To be submitted on Moodle with the group number by one student per group

1. **Video recording** of the project presentation as mp4 file, e.g.: `video_recording_group_01.mp4`
   (Ensure that the video does not exceed the max. allowed file size on Moodle. If the video is too large, split it and upload separate parts, e.g.: `video_recording_group_01_part_1.mp4`, `video_recording_group_01_part_2.mp4`, ...)
2. **Zip file** with all data and your Jupyter notebook(s), e.g.: `materials_group_01.zip`
3. **Presentation as PDF file** (e.g. `presentation_group_01.pdf`) with the following minimal structure:
   - Title and full name of each student
   - Introduction (background, problem, objective, research question)
   - Materials and methods (data collection method, content of data, exploratory data analysis methods, modelling methods)
   - Results & Discussion
   - Conclusions
   - Reflection on AI usage (which tools, for which tasks, which advantages and disadvantages you found, and where something did not work with the help of AI)

IMPORTANT!!! In the appendix of the PDF document of the presentation, include what you think you did to earn the points. Use screenshots and code snippets to clearly demonstrate this. Note that the final evaluation will be made by the lecturer, not exclusively based on your listing!

Submission of the final project (Deadline: January 10, 2027, 11:59 p.m.)
Due: Sunday, 11 January 2026, 11:59 PM

Mark as done

▶ Open all ▼ Close all

▼ Week 01 (15.09. - 18.09.)

Lecturers: Dr. Mario Gellrich; Dr. Alexander Walter

Slides GK Data Analytics Week 01 (Updated: 2026-09-02)

Material for exercises: https://github.com/mario-gellrich-zhaw/data_analytics

Connecting GitHub- with ZHAW Account (Step by Step Instruction)

▼ Week 02 (21.09. - 25.09.)

Lecturers: Dr. Mario Gellrich; Dr. Alexander Walter

Slides GK Data Analytics Week 02 (Updated: 2026-09-21)

Material for exercises: https://github.com/mario-gellrich-zhaw/data_analytics

▼ Week 03 (28.09. - 02.10.)

Lecturers: Dr. Mario Gellrich; Dr. Alexander Walter

Slides GK Data Analytics Week 03 (Updated: 2026-09-02)

Material for exercises: https://github.com/mario-gellrich-zhaw/data_analytics

MS Excel sverweis() / vlookup() Mark as done

Week 04 (05.10. - 09.10.)

Lecturers: Dr. Mario Gellrich; Dr. Alexander Walter

Slides GK Data Analytics Week 04 (Updated: 2026-09-02)

Material for exercises: https://github.com/mario-gellrich-zhaw/data_analytics

Text-to-SQL

Augmented Analytics with Python

Week 05 (12.10. - 16.10.)

Lecturers: Dr. Mario Gellrich; Dr. Alexander Walter

Slides GK Data Analytics Week 05 (Updated: 2026-09-02)

Material for exercises (part I): https://github.com/mario-gellrich-zhaw/data_analytics

Material for exercises (part II): https://github.com/mario-gellrich-zhaw/spatial_data_analysis

qgis tutorial 01

Install QGIS (long term stable version) on your local computer from here: https://www.qgis.org/download.

Note that the geodata are now available in your GitHub Fork -> Week 05, Folder 'Geodata'.

The Jupyter Notebook for the geocoding tasks is also available in your GitHub Fork -> Week 05.

qgis tutorial 02

qgis tutorial 03

cube.kml

Week 06 (19.10. - 23.10.)

Lecturers: Dr. Mario Gellrich; Dr. Alexander Walter

Slides GK Data Analytics Week 06 (Updated: 2026-09-02)

Material for exercises: https://github.com/mario-gellrich-zhaw/data_analytics

Week 07 (26.10. - 30.10.)

Lecturers: Dr. Mario Gellrich; Dr. Alexander Walter

Slides GK Data Analytics Week 07 (Updated: 2026-09-02)

Material for exercises: https://github.com/mario-gellrich-zhaw/data_analytics

Spearman Correlation (mandatory reading) Mark as done

Week 08 (02.11. - 06.11.)

Lecturers: Dr. Mario Gellrich; Dr. Alexander Walter

Slides GK Data Analytics Week 08 (Updated: 2026-09-02)

Material for exercises: https://github.com/mario-gellrich-zhaw/data_analytics

Random Forest including bagging and feature randomness (mandatory reading) Mark as done

Random Forest Classifier, the meaning of: bootstrap=True, max_samples=None, max_features='sqrt' (mandatory reading)

Week 09 (09.11. - 13.11.)

Lecturers: Dr. Mario Gellrich; Dr. Alexander Walter

Slides GK Data Analytics Week 09 (Updated: 2026-09-02)

Material for exercises: https://github.com/mario-gellrich-zhaw/data_analytics

Gini-Impurity calculation example (mandatory reading) Mark as done

Week 10 (16.11. - 20.11.)

Lecturers: Dr. Mario Gellrich; Dr. Alexander Walter

Slides GK Data Analytics Week 10 (Updated: 2026-09-02)

K-means (online)

Material for exercises: https://github.com/mario-gellrich-zhaw/data_analytics

Image segmentation based on k-means clustering (for voluntary use) Mark as done

Know What Your Neighbors Do: 3D Semantic Segmentation of Point Clouds (for voluntary use) Mark as done

Source of point cloud data as basis for semantic segmentation (for voluntary use) Mark as done

Week 11 (23.11. - 27.11.)

Lecturers: Dr. Mario Gellrich; Dr. Alexander Walter

Slides GK Data Analytics Week 11 (Updated: 2026-10-21)

Material for exercises: https://github.com/mario-gellrich-zhaw/data_analytics

Train a neural network using continuous bag-of-words - CBOW (for voluntary use) Mark as done

Week 12 (30.11. - 04.12.)

Lecturers: Dr. Mario Gellrich; Dr. Alexander Walter

Slides GK Data Analytics Week 12 (Updated: 2026-09-02)

Material for exercises: https://github.com/mario-gellrich-zhaw/data_analytics

Tensorflow Playground (for voluntary use)

Convolutional Neural Networks (for voluntary use) Mark as done

Visualization of Convolutional Neural Networks (for voluntary use) Mark as done

Week 13 (07.12. - 11.12.)

Lecturers: Dr. Mario Gellrich; Dr. Alexander Walter

Slides GK Data Analytics Week 13 (Updated: 2026-09-02)

Material for exercises: https://github.com/mario-gellrich-zhaw/data_analytics

Week 14 (14.12. - 18.12.)

Lecturers: Dr. Mario Gellrich; Dr. Alexander Walter

Infos Written Exam DA HS2026 (Updated: 2026-09-02)

Additional Material

PostgreSQL database via Docker (docker-compose.yml)

If you plan to integrate a PostgreSQL database in your project, here is an example.